

Cosmic origin of the Planck constant from a topological condensate phase transition: canonical structural formula $\hbar_{\text{TECT}} = c^3 a_{\text{BCC}}^2 / (16\pi G)$, four-layer GAP-1 verification programme, and the joint-event promotion theorem for the T4 $\rightarrow$ T6 composite-tier upgrade

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We present the canonical TECT structural derivation of the Planck constant $\hbar$ as a frozen-in adiabatic invariant of a Brazovskii-type body-centred-cubic (BCC) topological condensate. The construction combines the candidate condensate elastic modulus $\rho_{\text{cond}} = c^4 / (16\pi G a_{\text{BCC}}^2)$ (Math110-AddG, from the BCC continuum elastic theory) with the freeze-out time-scale $\tau_{\text{BCC}} = a_{\text{BCC}} / c_T$ at the transverse-phonon identification $c_T = c$ (Math110-AddH = Pillar 2 + Math115) to yield the canonical structural formula

$$\boxed{\hbar_{\text{TECT}} = \frac{c^3 a_{\text{BCC}}^2}{16\pi G}} \quad (\text{Math291 Theorem 291.1, Formula B canonical}).$$

The Math291 dimensional audit established Formula B as the unique dimensionally-admissible closed-form among three competing expressions in the prior GAP-1 archive (Formulas A and C were dimensionally incorrect, with the root cause identified as a single-symbol velocity-versus-time substitution defect in Math110-AddI Step 3, now audit-flagged for errata). At the canonical $a_{\text{BCC}} = 4\sqrt{\pi} \ell_P$ identification (the $R_{F5} = 1$ gate), Formula B numerically matches the observed $\hbar$ to $\sim 10^{-3}$ relative precision in code units. The Math299 Phase 3 closure consolidates the four-layer GAP-1 promotion pathway: (i) structural layer a_{BCC}^2 formula (Math291 + Math297, T6 PROVED CONDITIONAL with continuum-limit precision budget $|\delta a_{\text{BCC}} / a_{\text{BCC}}| < 5 \times 10^{-4}$ strict or $< 5 \times 10^{-3}$ relaxed); (ii) physical 1-loop matching $\gamma_{\hbar}^{(1)}(\mu) = \alpha_{\hbar}^{(1)} \ln(\mu / M_Z)$ with $\alpha_{\hbar}^{(1)} \approx -0.0706$ derived from 1-loop SM β -functions under the universal-embedding hypothesis (Math296); (iii) sector-decomposition framework c_1, c_2, c_3 for $U(1)_Y / SU(2)_L / SU(3)_c$ (Math298); (iv) composite GAP-1 tier (Math299 Theorem 299.1, T4 STRONG EVIDENCE currently, T6 PROVED CONDITIONAL upon the joint event $C_1 \wedge C_2 \wedge C_3$ on continuum precision + sector outcome + boundary-region absorption). The pre-registered F-GAP1 falsification gate $|\Delta \hbar / \hbar_{\text{obs}}| < 10^{-3}$ at the structural tier binds the matching residual; the verdict deadline is 2026-05-22 per Math312. The result establishes the canonical structural formula and sharpens the GAP-1 verification programme; full GAP-1 composite closure remains conditional on the joint-event verdict per Math299. The canonical GAP-1 composite tier as recorded in `Docs/status/TOE-FACT-SHEET.md` is currently **T4 STRONG EVIDENCE**.

I. INTRODUCTION AND THE STRUCTURAL QUESTION

The status of $\hbar$ as either an empirical input or a derived quantity has remained open across all major candidate Theory-of-Everything (TOE) frameworks. The Standard Model takes $\hbar$ as a postulated constant; the Standard formulation of superstring/M theory does not address its origin; Loop Quantum Gravity absorbs its functional content into the Immirzi parameter, which is itself an external input.

The TECT framework provides a candidate *structural* derivation: $\hbar$ emerges as the action accumulated by one BCC correlation cell during one Kibble–Zurek freeze-out period at the locked Brazovskii critical moment, with all the inputs of the candidate construction expressible in terms of c , G , and the BCC condensate lattice spacing a_{BCC} . The present paper reports the canonical structural derivation in its current form (Rev 4, 2026-05-03), synthesised from the Math97 + Math98-AddA + Math110-AddG/H/I + Math115 + Math145–147 + Math261 + Math291 + Math296 + Math297 + Math298 + Math299 + Math300 archive. The canonical GAP-1 composite tier remains **T4 STRONG EVIDENCE** per the canonical Docs/status/TOE-FACT-SHEET.md; promotion to T6 PROVED CONDITIONAL is gated by the Math299 Theorem 299.1 joint-event condition with verdict-shell deadline 2026-05-22 per Math312.

II. SETTING AND THE FOUR PRIMITIVES

Throughout we work in SI units. The TECT primitives are:

$$c \equiv \text{vacuum speed of light} = 2.99792458 \times 10^8 \text{ m/s}, \quad (1)$$

$$G \equiv \text{Newton constant} = 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \quad (2)$$

$$a_{\text{BCC}} \equiv \text{BCC condensate lattice spacing} \\ (\text{Math82-H continuum, Math110-AddB}), \quad (3)$$

$$c_T \equiv \text{transverse-phonon velocity} = c \quad (\text{Math110-AddH} = \text{Pillar 2} + \text{Math115}). \quad (4)$$

The candidate condensate mass-energy density is identified, by the Math110-AddG closure of the BCC continuum elastic theory, as

$$\rho_{\text{cond}} = \frac{c^4}{16\pi G a_{\text{BCC}}^2}. \quad (5)$$

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This identification is dimensionally clean ($[\rho_{\text{cond}}] = \text{kg m}^{-3}$) and is verified by direct substitution. The transverse-phonon identification $c_T = c$ is the load-bearing input from Pillar 2 (kinematic Lorentz consistency, Math115); the candidate emergent equivalence between transverse condensate kinematics and the speed of light is the Pillar 2 / Pillar 3 joint statement, treated separately in the companion Paper 02 + Paper 03.

III. MATH97 + MATH98-ADDA: BRAZOVSKII UNIVERSALITY AND KIBBLE–ZUREK τ -POWER-TAIL DERIVATION

The Brazovskii free energy

$$\mathcal{F}[\Psi] = \int d^3x \left[\frac{1}{2} |\nabla \Psi|^2 + \frac{1}{2} \mu^2 |\Psi|^2 + \frac{\gamma}{2} (\Delta + q_0^2)^2 |\Psi|^2 + \frac{\lambda}{4} |\Psi|^4 \right] \quad (6)$$

governs the candidate primordial BCC condensate from its high-temperature symmetric phase through Kibble–Zurek freeze-out to the BCC-locked broken phase. The Math97 conditional universality-class membership programme (companion auxiliary paper Auxiliary-01) provides the candidate framework; the Math98-AddA τ -power-tail derivation gives the explicit Kibble–Zurek freeze-out scaling under the Brazovskii first-order locking.

Proposition 1 (Math98-AddA Kibble–Zurek freeze-out scaling). *At the candidate Brazovskii first-order critical moment, the BCC correlation length ξ and the freeze-out time τ_{freeze} satisfy*

$$\xi \rightarrow a_{\text{BCC}}, \quad \tau_{\text{freeze}} \rightarrow \tau_{\text{BCC}} \equiv \frac{a_{\text{BCC}}}{c_T}. \quad (7)$$

The correlation cell volume is $V_\xi = \xi^3 = a_{\text{BCC}}^3$ at the critical moment.

IV. MATH291 CANONICAL STRUCTURAL FORMULA AND THE DIMENSIONAL AUDIT

The Math291 Phase 1 canonical-hygiene errata note resolves a three-formula discrepancy in the GAP-1 archive prior to Turn 61. We state the canonical result as follows.

Theorem 2 (Math291 Theorem 291.1, canonical structural $\hbar_{\text{T E C T}}$ formula — Formula B). *The Kibble–Zurek action accumulated by one BCC correlation cell during one freeze-out period at the candidate Brazovskii critical moment is*

$$S_{\text{KZ}}^{\text{physical}} = \rho_{\text{cond}} \cdot a_{\text{BCC}}^3 \cdot \tau_{\text{BCC}}. \quad (8)$$

Substituting $\rho_{\text{cond}} = c^4/(16\pi G a_{\text{BCC}}^2)$ (Eq. 5) and $\tau_{\text{BCC}} = a_{\text{BCC}}/c_T$ (Proposition 1) at the Math110-AddH transverse-phonon identification $c_T = c$ yields

$$\boxed{\hbar_{\text{TECT}}^{\text{B}} = \frac{c^3 a_{\text{BCC}}^2}{16\pi G}}. \quad (9)$$

This expression has SI dimensions $[\hbar^{\text{B}}] = (\text{L T}^{-1})^3 \cdot \text{L}^2/(\text{L}^3 \text{M}^{-1} \text{T}^{-2}) = \text{ML}^2 \text{T}^{-1} = \text{J} \cdot \text{s}$ (action) ✓ and is the unique dimensionally-admissible closed-form among the three expressions appearing in the historical GAP-1 archive.

Remark 3 (Math291 errata: Formulas A and C are dimensionally inadmissible). The historical Math110-AddI Eq. (1.4) Formula A $\hbar^{\text{A}} = c^5 a_{\text{BCC}}/(16\pi G)$ and the Math283 §3 line-161 Formula C $\hbar^{\text{C}} = c^3 a_{\text{BCC}}/(16\pi G)$ both fail the SI dimensional check (giving power-times-length and momentum, respectively, neither of which is action). The root-cause of Formula A is identified in Math291 §root as a single-symbol substitution defect in Math110-AddI Step 3: the velocity c_T was used in place of the time $\tau_{\text{BCC}} = a_{\text{BCC}}/c_T$, inflating the result by the multiplicative factor c^2/a_{BCC} . Formula C is the downstream further misprint dropping one further factor of a_{BCC} . Both are audit-flagged for errata in the canonical archive; no status row of any pillar changes (Math286 already recorded the canonical $\text{H1.G1} = \text{T6 PROVED CONDITIONAL}$ with Formula B silently in place).

A. Numerical evaluation at the canonical $R_{F5} = 1$ identification

At the canonical Math110-AddI / Math110-AddB BCC primitive

$$a_{\text{BCC}} = 4\sqrt{\pi} \ell_P, \quad \ell_P = \sqrt{\hbar G/c^3}, \quad (10)$$

direct substitution into Eq. (9) gives $\hbar_{\text{TECT}}^{\text{B}} \approx 1.055 \times 10^{-34} \text{ J} \cdot \text{s}$, matching the CODATA-recommended $\hbar_{\text{obs}}$ to relative precision $\sim 10^{-3}$ in TECT code units. The $a_{\text{BCC}} = 4\sqrt{\pi} \ell_P$ identification is the $R_{F5} = 1$ gate of Math110-AddI; in the canonical structural reading, this gate reduces the empirical input list from $\{c, G, \hbar\}$ to $\{c, G\}$ plus one cosmological-cooling axiom (Math195 A2-axiom-reducibility programme, see Top-impact paper TI-3 for the *conditional* character of this reduction).

V. FOUR-LAYER GAP-1 VERIFICATION PROGRAMME (MATH296–MATH299)

The Math299 Phase 3 closure note consolidates the GAP-1 verification programme into a four-layer cascade:

TABLE I. The four-layer GAP-1 closure cascade and Phase 1+2+3 cited notes (Math299 §setting Table).

Layer	Closure note	Tier (current)
Structural (a^2 formula)	Math291 + Math297	T6 + budget
Physical 1-loop	Math296 + Math298	T4 (ansatz)
Physical 2-loop+	Tasks #147/#148	T1 (open)
Composite GAP-1	Math299 (this)	T4 (min rule)

A. Math296 1-loop matching functional

Theorem 4 (Math296 1-loop matching ansatz). *Under the universal-embedding hypothesis (TECT $\hbar$ couples to the SM beta-function spectrum through a single universal coefficient), the 1-loop matching functional takes the linear-log form*

$$\gamma_h^{(1)}(\mu) = \alpha_h^{(1)} \ln(\mu/M_Z), \quad \alpha_h^{(1)} \approx -0.0706, \quad (11)$$

derived from the standard 1-loop Standard-Model β -functions (Machacek–Vaughn 1983) under the BCC-locked structural primitives. The form is derived (not postulated) from first-principles SM 1-loop matching.

B. Math297 continuum-limit precision budget

Theorem 5 (Math297 continuum-limit error budget). *For Theorem 2 (Formula B) to satisfy the F-GAP1 structural-tier binding $|\Delta\hbar/\hbar_{\text{obs}}| < 10^{-3}$, the Math82-H continuum extrapolation on the canonical $N \in \{16, 32, 64\}$ Richardson ladder must deliver a_{BCC} to relative precision*

$$\left| \frac{\delta a_{\text{BCC}}}{a_{\text{BCC}}} \right| < \begin{cases} 5 \times 10^{-4} & (\text{strict, Outcome A}), \\ 5 \times 10^{-3} & (\text{relaxed, Outcome B with 1-loop matching absorbing the residual}). \end{cases} \quad (12)$$

The strict outcome closes the structural tier directly; the relaxed outcome closes the structural tier conditionally, with the residual absorbed by Math296 1-loop matching.

C. Math298 sector decomposition

Proposition 6 (Math298 sector decomposition framework). *The empirical 1-loop residual is decomposed as $\delta_{\text{emp}}(\mu) = c_1 b_1(\mu) + c_2 b_2(\mu) + c_3 b_3(\mu)$ where $b_i(\mu)$ are the SM 1-loop β -function*

contributions of $U(1)_Y/SU(2)_L/SU(3)_c$ respectively. The fit selects between Outcome U (universal embedding $c_1 = c_2 = c_3$) and Outcome $S_{i^*=3}$ (QCD-dominant, $c_3 \gg c_1, c_2$) which validates or refines the universal-embedding hypothesis of Math296.

D. Math299 joint-event $T4 \rightarrow T6$ promotion theorem

Theorem 7 (Math299 GAP-1 $T4 \rightarrow T6$ promotion condition). *The GAP-1 composite tier (currently $T4$ STRONG EVIDENCE per Math286 H1 row) promotes to $T6$ PROVED CONDITIONAL upon joint satisfaction of all three conditions:*

1. **(C1) Continuum-limit precision:** *Math82-H Richardson extrapolation on $N \in \{16, 32, 64\}$ delivers $|\delta a_{\text{BCC}}/a_{\text{BCC}}| \leq 5 \times 10^{-3}$ (Outcome A or B of F-Math297-aBCC-precision).*
2. **(C2) Sector-decomposition outcome:** *Tasks #147/#148 empirical $\delta_{\text{emp}}(\mu)$ fits to either Outcome U or Outcome $S_{i^*=3}$ of F-Math298-Sector.*
3. **(C3) Boundary-region absorption:** *at the matched μ^* effective transition scale, the 1-loop ansatz residual satisfies $|\Delta\hbar(\mu^*)/\hbar_{\text{obs}}| < 10^{-3}$ (F-GAP1 binding).*

The promotion fires when $C_1 \wedge C_2 \wedge C_3 = \top$. Failure of any one condition leaves GAP-1 at $T4$ with the corresponding sub-pathway flagged as the residual obstruction.

Remark 8 (Verdict-shell schedule per Math312). The verdict on the joint event $C_1 \wedge C_2 \wedge C_3$ is scheduled to fire on the F-GAP1 deadline 2026-05-22 per the Math312 verdict-consumption shell. The current paper records the $T4$ STRONG EVIDENCE composite tier per the canonical Docs/status/TOE-FACT-SHEET.md; the $T4 \rightarrow T6$ promotion is gated by the verdict-shell outcome and does not occur within the present paper's scope.

VI. HONEST SCOPE AND OPEN PROGRAMME ITEMS

a. What the present paper establishes. At the canonical **T4 STRONG EVIDENCE** tier per the canonical Docs/status/TOE-FACT-SHEET.md, the present paper reports: (i) the canonical structural formula $\hbar_{\text{T ECT}} = c^3 a_{\text{BCC}}^2 / (16\pi G)$ (Theorem 2, Math291 Formula B), with the dimensional audit of Remark 3 establishing it as the unique dimensionally-admissible closed-form among the historical candidates; (ii) the four-layer GAP-1 verification programme (§V, Math296 + Math297 + Math298 + Math299); (iii) the explicit $T4 \rightarrow T6$ promotion condition as the joint event $C_1 \wedge C_2 \wedge C_3$

of Theorem 7; (iv) the pre-registered F-GAP1 falsification gate $|\Delta\hbar/\hbar_{\text{obs}}| < 10^{-3}$ at the structural tier with verdict deadline 2026-05-22 per Math312.

b. What the present paper does NOT establish. The paper does NOT establish: (a) the F-GAP1 verdict itself, which awaits the joint event $C_1 \wedge C_2 \wedge C_3$ verification at the Math312 verdict-shell deadline; (b) the strict structural closure (Outcome A of Math297), which depends on the Math82-H Richardson extrapolation reaching $|\delta a_{\text{BCC}}/a_{\text{BCC}}| < 5 \times 10^{-4}$ on the $N \in \{16, 32, 64\}$ ladder; (c) the unconditional T6 PROVED CONDITIONAL upgrade of GAP-1 composite tier (Math299 Theorem 7); (d) the universal-embedding hypothesis of Math296, which is empirically distinguishable from the QCD-dominant alternative via the Math298 sector decomposition; (e) the 2-loop and higher physical matching, which is at T1 OPEN in the Math299 cascade (Tasks #147/#148).

c. Implications for downstream pillars and Stage-2. The Pillar 10 (Math79-AddB classical no-go theorem + Math110-AddI Kibble–Zurek resolution) consumes the canonical Formula B as the structural anchor. The GAP-1 composite tier remains T4 STRONG EVIDENCE pending the joint-event verdict; the Stage-2 composite tier remains T3 per Math307 + Math310 + Math311 pending the joint event over F-GAP1 / F-GAP4 / F-Pillar6 across the Math312 verdict-shell schedule.

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